Not asked or not answered? Missingness provenance in youth suicide-attempt prediction using the combined Youth Risk Behavior Survey, 2015–2023

Selorm Buaka PhD Student, Applied Statistics and Research Methods, University of Northern Colorado, Greeley, Colorado, USA ORCID 0009-0005-3991-4859 · othnielbuaka@gmail.com

Keywords: missing data; survey methodology; Youth Risk Behavior Survey; suicide attempt; prediction models; adolescent health; public health surveillance

Abstract

Background. Pooled multi-site surveys mix several reasons a predictor can be absent: a jurisdiction did not ask the item, the item was not on that year's questionnaire, or a student did not answer. Prediction models usually treat all of these as one missing value.

Methods. We used the CDC Youth Risk Behavior Survey combined high-school data, 2015–2023 (1,008,297 records; 250 site-year surveys). Each predictor cell was coded by provenance using only what the released data identify. Ten models predicted past-year suicide attempt (684,688 students) and injurious suicide attempt (504,707 students) from 15 behavioural items and demographics. Conventional handling (complete-case, mean imputation, missing indicators, chained-equations imputation, gradient boosting) was compared with provenance-aware indicators and pattern submodels. Evaluation used held-out school clusters, survey weights and a cluster bootstrap. A structure-only model and 12 sensitivity analyses tested whether gains reflected survey recognition. BRFSS 2023 adverse childhood experience items served as an adult contrast.

Results. Weighted prevalence was 8.8–10.7% for any attempt and 3.1–3.4% for injurious attempt. Of absent predictor cells, most came from surveys not asking the item (10.5% of all cells) rather than missingness where asked (3.5%). Provenance-split indicators improved AUROC over a single missing indicator by 0.0037 (95% CI 0.0019–0.0057) for any attempt and 0.0057 (0.0023–0.0090) for injurious attempt; the best model reached 0.853 and 0.849. A site-year-and-pattern-only model reached 0.55. The gain disappeared in the national survey, where sites do not omit items, and grew when whole jurisdictions were held out (0.0054 and 0.0104). In BRFSS, provenance handling made no difference.

Conclusions. Distinguishing "not asked" from "not answered" gives small, consistent and transportable gains in pooled youth-survey prediction that are not explained by survey recognition. Pooled-survey analyses should code the source of absence.

1. Introduction

The Youth Risk Behavior Surveillance System (YRBSS) is the main U.S. source of school-based data on adolescent suicidal behavior. CDC's combined high-school data set pools the national survey with state and large urban district surveys so that multi-site, multi-year analyses are possible [1].

In the pooled file, a blank predictor can mean different things. A jurisdiction may have dropped the question from its questionnaire. The question may not have been on that year's standard questionnaire. The student may have skipped it, or CDC may have set an inconsistent answer to missing. CDC states that, at the individual level, these mechanisms cannot be separated in the released data [2]. The 2023 move to tablet administration added skip patterns, but legitimately skipped questions are coded to their implied response rather than left missing [2].

Prediction studies usually treat every blank as one kind of missing value. In a pooled file this

has two risks. A model can lose information that differs by source of absence. It can also learn which survey a student came from, because survey content and outcome prevalence both vary by site and year. Work on missing data for prediction shows that the right handling depends on what is available at deployment [3, 4], and pattern-based models can avoid imputing values that were never collected [5].

Machine-learning prediction of adolescent suicide attempts from school-based risk-behaviour surveys is established, much of it in the Korea Youth Risk Behavior Survey [6, 7]. Nonresponse has also been studied as an outcome in its own right in YRBS, for the suicide-attempt item [8] and the sexual identity item [9]. We did not find work that separates survey-level omission from student-level nonresponse when building prediction models from the pooled YRBS file.

We asked three questions using 2015–2023 YRBS data:

1. How is absent predictor information distributed across sources that the released data can identify?
2. Does provenance-aware handling change discrimination and calibration for predicting past-year suicide attempt and injurious suicide attempt, compared with conventional handling?
3. Is any gain explained by survey structure (site and year) rather than student-level information?

The Behavioral Risk Factor Surveillance System (BRFSS) adverse childhood experiences (ACE) module served as a contrasting adult test bed, where state participation, questionnaire versions and in-module breakoff create different kinds of absence.

2. Methods

The analysis protocol, hypotheses and decision log were written before any test-set scoring; the one post hoc change is reported in Section 2.7. Reporting follows the TRIPOD+AI statement where it applies [10].

2.1 Data

We used the CDC YRBS 2023 combined high-school data set (1991–2023 release), restricted to 2015–2023 by pre-stated availability rules: each outcome asked in at least half of a year's surveys, and at least 80% of predictor concepts on that year's questionnaire. All five cycles from 2015 to 2023 met both rules. The file held 1,008,297 student records from 250 site-year surveys. The primary population was state and district surveys. The national survey was analysed separately and never pooled with site surveys in a model, because the two represent different target populations.

2.2 Outcomes

- **Any past-year suicide attempt**: one or more attempts in the past 12 months (Q29; CDC QN29 definition).
- **Injurious suicide attempt**: an attempt resulting in injury, poisoning or overdose treated by a doctor or nurse, among all respondents including those who did not attempt (Q30; CDC QN30 definition).

Records with a missing outcome, or from surveys that did not ask the outcome item, were excluded (Table 1). The primary cohorts held 684,688 students in 235 surveys (any attempt) and 504,707 students in 211 surveys (injurious attempt).

Table 1. Cohort flow, state and district surveys, 2015–2023.

Step	Any attempt: students	Any attempt: surveys	Injurious: students	Injurious: surveys
Records in the combined file, 1991–2023 release	1,008,297	250	1,008,297	250
Survey years 2015–2023	1,008,297	250	1,008,297	250
State and district surveys (national excluded)	926,896	245	926,896	245
Weighted (representative) surveys	926,896	245	926,896	245
Survey asked the outcome item	766,807	235	590,937	211
Valid outcome response (analytic cohort)	684,688	235	504,707	211

2.3 Predictors

Fifteen CDC dichotomous indicators covered sadness or hopelessness, bullying at school and online, physical fighting, carrying a weapon at school, forced sex, physical dating violence, skipping school for safety, current alcohol, marijuana, vaping and cigarette use, prescription pain medicine misuse, sleep and grades (Appendix Table A1). Variables were matched to concepts from CDC's SAS labels. Demographics (age, sex, grade, race and ethnicity, sexual identity) entered with an explicit missing level. Suicidal ideation and planning were excluded from primary models as same-construct items. Site and year identifiers were excluded from individual-level models.

2.4 Missingness provenance

Each predictor cell received exactly one code, restricted to what the released data can identify (Table 2).

Table 2. Missingness-provenance codes.

Code	Meaning	Sources it contains
OBS	Observed response	Answered items; skip-implied values from electronic administration
OBS_NONSUB	Observed, structurally not applicable	QN22 missing because the student did not date (Q22 = "did not date")
NA_SITE	Not asked by this survey	Item missing for every record in the site-year survey but present in other surveys that year
NA_YEAR	Not on that year's questionnaire	Item missing in every survey that year
NA_UNRES	Missing where asked	Item nonresponse and CDC edit-to-missing, which cannot be separated

Skip-implied values were treated as observed, in line with CDC coding. Quality-control exclusions are not visible in the release and are a selection limitation.

2.5 Comparators

Ten models were compared (Table 3).

Table 3. Comparators.

Model	Role	Missing-data handling
CC_LR	Conventional	Complete-case logistic regression
MEAN_LR	Conventional	Training-mean imputation
MIND_LR	Conventional	Mean imputation + one missing indicator per item
MICE_LR	Conventional	Chained-equations imputation (m = 5, outcome excluded), predictions averaged
HGB_NATIVE	Conventional	Histogram gradient boosting with native missing-value routing
PROV_LR	Provenance-aware	Mean imputation + separate not-asked, unresolved and not-applicable indicators
PROV_HGB	Provenance-aware	Native routing + provenance indicators
PATTERN_LR	Structure-aware	One logistic model per survey availability pattern [5]
STRUCT_ONLY	Diagnostic	Site, year and availability pattern only; no student answers
DEMO_LR	Diagnostic	Demographics only

All preprocessing was fit on training rows only. Hyperparameters were fixed in advance (logistic: L2, C = 1; boosting: learning rate 0.05, up to 500 iterations with early stopping, 31 leaves, at least 50 samples per leaf). The pre-specified primary contrasts were PROV_LR vs MIND_LR, PATTERN_LR vs MEAN_LR, PROV_HGB vs HGB_NATIVE and PROV_LR vs MICE_LR.

2.6 Splits

Within each site-year survey, primary sampling units (PSUs) were assigned 60/20/20 to training, validation and test sets with a fixed seed, so no school cluster spans training and test. Splits were hashed and frozen. For any attempt this gave 411,571 training, 136,819 validation and 136,298 test records; for injurious attempt, 305,003, 100,701 and 99,003. Alternative structures held out whole jurisdictions (leave-sites-out) or the latest cycle (temporal).

2.7 Survey design and evaluation

Prevalence was estimated with survey weights and Taylor-linearized standard errors, with strata and PSUs made unique per site-year. Models were trained unweighted; because evidence on weighting in gradient boosting is mixed [11], weighted training was a sensitivity analysis. Performance was estimated on the test set both unweighted and survey-weighted; weighted results are reported unless stated.

Metrics were AUROC, PR-AUC, Brier score, calibration intercept and slope, and sensitivity, specificity, PPV and NPV at three thresholds chosen on the validation set (80% sensitivity; top 10% and top 5% of predicted risk). Subgroups were reported when they had at least 30 events.

Confidence intervals came from a bootstrap that resampled PSUs with replacement within each site-year survey, with the fitted model held fixed (1,000 replicates for every analysis). Method contrasts used paired draws. **Post hoc change:** the protocol specified the surveys' own design strata. After the first run, 1,874 of 2,908 design strata (64%) held a single test-set PSU, which contributes no bootstrap variance. We therefore resampled within whole surveys (1 of 235 surveys had a single test PSU). This is conservative and leaves point estimates unchanged.

2.8 Sensitivity analyses

Twelve pre-specified scenarios were run for both outcomes: leave-sites-out; temporal; national survey only; excluding 2021; excluding 2023; 2023 only; dropping items not asked in more than 30% of surveys; a 95% threshold for “not asked”; adding ideation and planning; adding site and year fixed effects; weighted training; and including unweighted surveys.

2.9 BRFSS test bed

We used the 2023 BRFSS main file and questionnaire versions 1–3 [13]. The outcome was ever being told of a depressive disorder (ADDEPEV3). Predictors were the 11 ACE items, whose module nonresponse has been described previously [12]. Provenance codes were: state did not field the module; module fielded only on a version this respondent did not receive; breakoff within the module; other blank; don’t know or refused; and “parents not married” (observed, not applicable). Models were evaluated within states and leaving states out.

2.10 Software

Analyses used Python with pandas, NumPy and scikit-learn. Fixed-width files were read with layouts parsed from CDC’s SAS input programs. The full pipeline ran on a laptop; scripts for a Slurm cluster are included in the code release.

3. Results

3.1 Cohort and outcome prevalence

In state and district surveys, weighted prevalence of any past-year suicide attempt ranged from 8.8% (2015) to 10.7% (2021). Injurious attempt ranged from 3.1% (2019) to 3.4% (2021), about one student in 30 (Appendix Table A2). In the national survey, 2023 weighted prevalence of any attempt was 9.5% (95% CI 8.4–10.6%), matching CDC’s published estimate.

3.2 Where predictor information was absent

Across the 15 predictors, 82.0% of cells were observed. Not asked by the survey was the largest absent class (10.5% of cells), followed by unresolved missingness where asked (3.5%), structurally not applicable (2.5%, all from dating violence) and not on the year’s questionnaire (1.5%, all from prescription pain medicine in early cycles) (Figure 1; Appendix Table A1). Site omission was all-or-nothing: no item was between 95% and 99.99% missing in any survey, so the 95% sensitivity rule labelled the same cells as the primary rule.

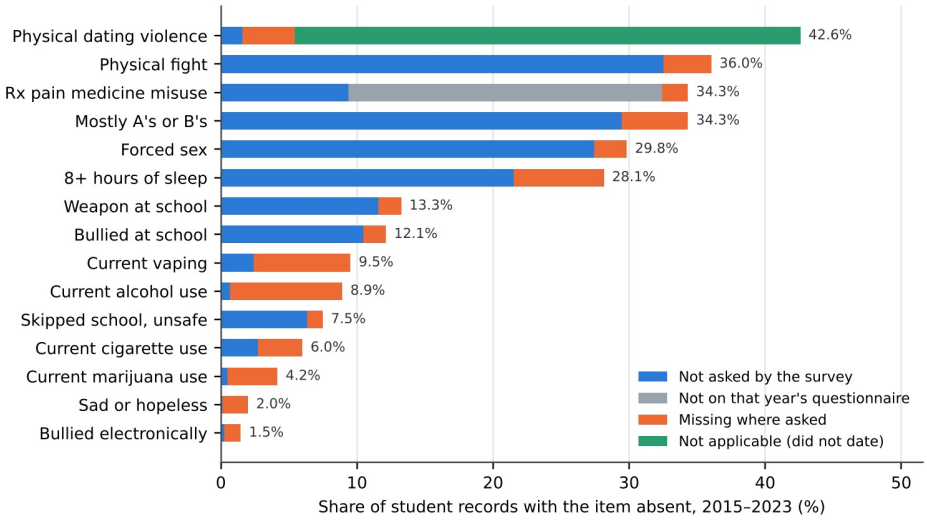


Figure 1. Share of student records with each predictor absent, by provenance, all 1,008,297 records 2015–2023 (national, state and district surveys). Most absence comes from surveys that did not ask the item, not from students who skipped it.

Complete-case analysis was therefore impractical. Only 8.6% of test-set students in the any-attempt cohort, and 11.4% in the injurious cohort, had all 15 items observed.

3.3 Primary comparison

Test sets held 136,298 students (any attempt) and 99,003 students (injurious attempt). Table 4 and Figure 2 show survey-weighted performance.

Table 4. Survey-weighted test-set discrimination, primary analysis (95% CI, 1,000 PSU bootstrap replicates).

	Any past-year attempt			Injurious attempt		
Model	**AUROC**	**PR-AUC**	**PPV @80% sens.**	**AUROC**	**PR-AUC**	**PPV @80% sens.**
PROV_HGB	0.853 (0.845–0.861)	0.458 (0.436–0.478)	0.251 (0.239–0.263)	0.849 (0.833–0.864)	0.231 (0.207–0.260)	0.085 (0.078–0.091)
HGB_NATIVE	0.849 (0.841–0.857)	0.451 (0.430–0.472)	0.243 (0.232–0.255)	0.842 (0.825–0.856)	0.223 (0.199–0.254)	0.078 (0.072–0.083)
PROV_LR	0.846 (0.838–0.854)	0.440 (0.419–0.460)	0.244 (0.233–0.256)	0.843 (0.826–0.859)	0.214 (0.192–0.239)	0.078 (0.072–0.083)
MIND_LR	0.843 (0.833–0.851)	0.435 (0.413–0.457)	0.238 (0.227–0.250)	0.838 (0.821–0.852)	0.215 (0.191–0.244)	0.072 (0.067–0.077)
PATTERN_LR	0.842 (0.834–0.850)	0.418 (0.400–0.437)	0.235 (0.224–0.245)	0.823 (0.802–0.841)	0.200 (0.176–0.227)	0.070 (0.065–0.075)
MICE_LR	0.841 (0.832–0.849)	0.425 (0.406–0.443)	0.237 (0.226–0.248)	0.828 (0.808–0.846)	0.207 (0.183–0.234)	0.068 (0.063–0.072)
MEAN_LR	0.839 (0.831–0.847)	0.414 (0.396–0.433)	0.238 (0.227–0.249)	0.823 (0.802–0.841)	0.195 (0.172–0.222)	0.069 (0.064–0.074)
CC_LR†	0.815 (0.786–0.844)	0.376 (0.333–0.422)	0.202 (0.181–0.223)	0.801 (0.762–0.844)	0.210 (0.160–0.281)	0.049 (0.039–0.059)
DEMO_LR	0.650 (0.638–0.661)	0.190 (0.176–0.205)	0.119 (0.113–0.125)	0.619 (0.595–0.644)	0.052 (0.046–0.061)	0.034 (0.032–0.037)
STRUCT_ONLY	0.552 (0.536–0.569)	0.122 (0.113–0.134)	0.106 (0.099–0.112)	0.557 (0.539–0.577)	0.038 (0.034–0.044)	0.031 (0.029–0.034)

PPV @80% sens.: positive predictive value at the threshold giving 80% sensitivity on the validation set. † Complete-case analysis scores only students with all items observed (8.6% and 11.4% of the test sets), so its metrics describe a different, smaller population.

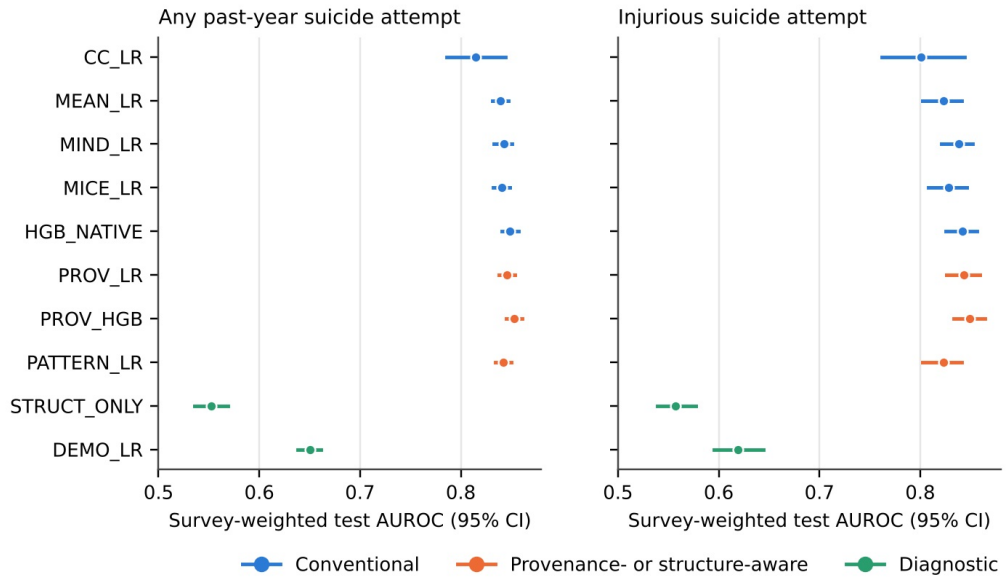


Figure 2. Survey-weighted test-set AUROC with 95% PSU-bootstrap intervals, primary analysis. The diagnostic STRUCT_ONLY model uses only site, year and which items the survey asked.

All four pre-specified contrasts favoured the provenance- or structure-aware model for any attempt. AUROC differences were PROV_LR vs MIND_LR +0.0037 (95% CI +0.0019 to +0.0057), PATTERN_LR vs MEAN_LR +0.0027 (+0.0015 to +0.0040), PROV_HGB vs HGB_NATIVE +0.0039 (+0.0024 to +0.0053) and PROV_LR vs MICE_LR +0.0057 (+0.0030 to +0.0085). For injurious attempt the corresponding differences were +0.0057 (+0.0023 to +0.0090), +0.0003 (−0.0021 to +0.0024), +0.0076 (+0.0034 to +0.0117) and +0.0156 (+0.0097 to +0.0223).

Calibration slopes for the imputation, indicator, provenance and boosting models were close to 1 (0.96–1.04; Table 5, Figure 4). For injurious attempt, calibration intercepts were negative for most models (for example −0.091 for MIND_LR and −0.074 for PROV_LR), indicating slight overprediction in the weighted population.

Table 5. Survey-weighted Brier score and calibration intercept and slope, primary analysis (95% CI).

	Any past-year attempt			Injurious attempt		
Model	**Brier**	**Intercept**	**Slope**	**Brier**	**Intercept**	**Slope**
PROV_HGB	0.069 (0.066–0.072)	0.020 (-0.032–0.078)	0.999 (0.957–1.038)	0.025 (0.023–0.027)	-0.081 (-0.160–-0.004)	0.993 (0.936–1.052)
HGB_NATIVE	0.070 (0.067–0.073)	0.012 (-0.040–0.069)	0.992 (0.950–1.033)	0.025 (0.023–0.027)	-0.125 (-0.202–-0.051)	0.971 (0.913–1.031)
PROV_LR	0.071 (0.067–0.074)	0.048 (-0.006–0.106)	1.004 (0.964–1.043)	0.025 (0.024–0.027)	-0.074 (-0.155–0.003)	0.964 (0.906–1.021)
MIND_LR	0.071 (0.068–0.074)	0.017 (-0.036–0.073)	1.001 (0.960–1.042)	0.025 (0.024–0.027)	-0.091 (-0.173–-0.017)	0.968 (0.908–1.030)
PATTERN_LR	0.072 (0.069–0.075)	0.050 (-0.001–0.105)	1.028 (0.991–1.066)	0.026 (0.024–0.027)	-0.113 (-0.195–-0.040)	0.992 (0.928–1.056)
MICE_LR	0.071 (0.068–0.075)	0.041 (-0.011–0.095)	1.043 (1.006–1.081)	0.025 (0.024–0.027)	-0.050 (-0.131–0.024)	0.983 (0.918–1.043)
MEAN_LR	0.072 (0.069–0.075)	0.007 (-0.045–0.060)	1.030 (0.991–1.067)	0.026 (0.024–0.027)	-0.064 (-0.145–0.009)	0.962 (0.898–1.023)
CC_LR	0.069 (0.063–0.074)	0.006 (-0.100–0.103)	0.934 (0.834–1.050)	0.021 (0.018–0.025)	-0.103 (-0.315–0.085)	0.924 (0.798–1.068)

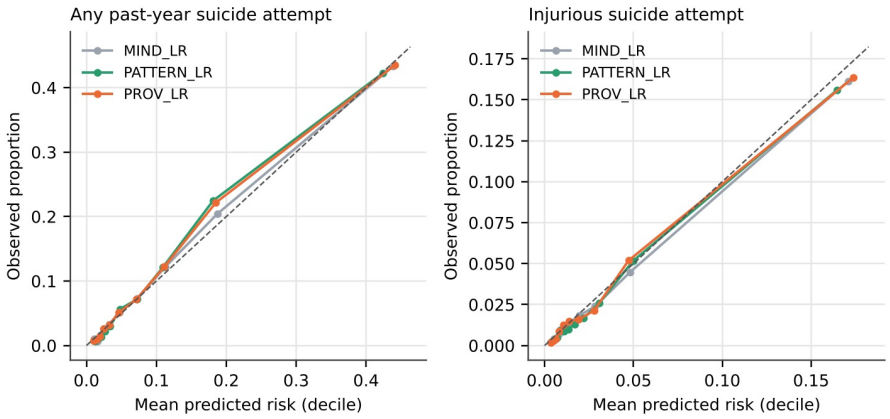


Figure 4. Survey-weighted calibration by decile of predicted risk, primary test sets. The dashed line marks perfect calibration.

3.4 Survey structure versus student information

A model using only site, year and which questions the survey asked reached AUROC 0.552 (any attempt) and 0.557 (injurious), below demographics alone. Provenance-aware gains therefore did not come from recognising surveys.

Three sensitivity analyses sharpen this (Figure 3; Appendix Tables A3 and A4):

- **National survey only**, where site omission does not occur: the PROV_LR vs MIND_LR gain disappeared for any attempt (−0.0004, −0.0033 to +0.0018).
- **Whole jurisdictions held out**: the gain was larger, +0.0054 (+0.0022 to +0.0086) for any attempt and +0.0104 (+0.0032 to +0.0178) for injurious attempt.
- **Site and year fixed effects added**: the any-attempt gain held (+0.0035), but the injurious gain shrank to +0.0018 (−0.0021 to +0.0054), so part of the injurious advantage tracks survey structure.

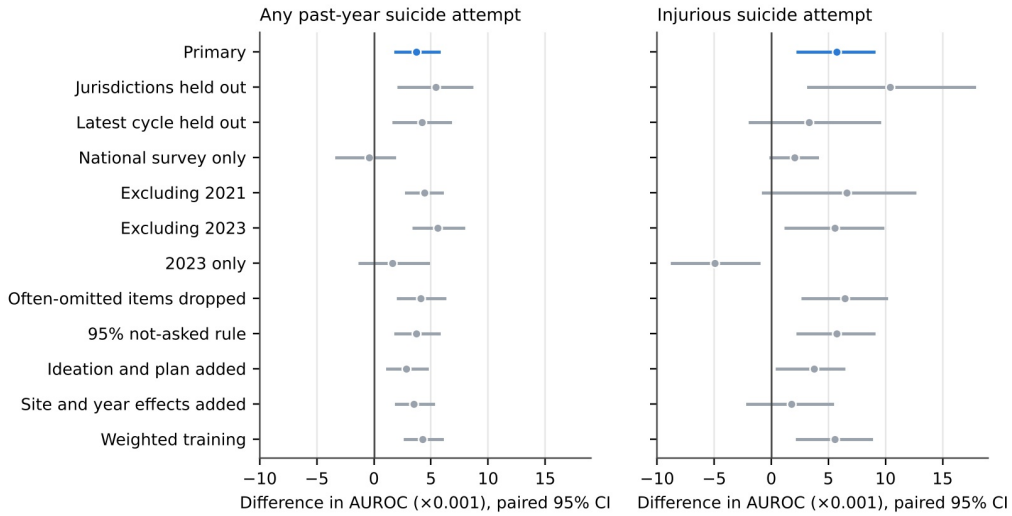


Figure 3. PROV_LR minus MIND_LR difference in survey-weighted AUROC (×0.001) across the pre-specified scenarios, paired 95% CI, 1,000 replicates. Right of zero favours provenance-split indicators. The national survey, where sites do not omit items, shows no gain.

3.5 Other sensitivity analyses

Excluding 2021, excluding 2023, dropping items not asked in over 30% of surveys and weighted training all left the direction and size of the provenance contrasts similar. In 2023 alone, the PROV_LR vs MIND_LR difference for injurious attempt was negative (−0.0049,

−0.0087 to −0.0011), and excluding 2021 widened the injurious interval to include zero (+0.0066, −0.0007 to +0.0125). Adding ideation and planning raised AUROC to 0.905–0.922 for any attempt and the provenance contrasts stayed positive. Every survey in the window carried a weight, so the planned “include unweighted surveys” scenario was identical to the primary analysis.

3.6 Subgroups

Discrimination was lower for students identifying as gay or lesbian (PROV_LR AUROC 0.780, 0.739–0.838) or questioning (0.761, 0.696–0.823) than heterosexual students (0.851, 0.825–0.874) for any attempt. It was also lower for Native Hawaiian or other Pacific Islander students (0.779, 0.703–0.851) and Black students (0.802, 0.776–0.834) than White students (0.862, 0.851–0.874) (Appendix Table A5). Sexual identity was not asked in every survey, so these subgroups come from a subset of sites.

3.7 BRFSS contrasting test bed

The BRFSS cohort held 430,736 adults; weighted prevalence of ever-diagnosed depressive disorder was 20.2% (19.9–20.4%). ACE items were absent for 76.9% of respondents because their state did not field the module, 7.1% because their state fielded it only on a split questionnaire version they did not receive (Ohio and Maryland), and 3.1% who were in module states but answered no ACE item. Don’t-know or refused answers were 0.2–0.4% per item and in-module breakoff 0.02–0.14%. Provenance handling made almost no difference: within-state AUROC was 0.683 for MIND_LR and 0.684 for PROV_LR, and leave-states-out AUROC was 0.709 for both (Appendix Table A6). Indiana is listed by CDC as using the module, but none of the four public 2023 files (main and versions 1–3) holds Indiana ACE responses.

4. Discussion

In about one million YRBS records, most absent predictor information came from surveys that did not ask an item, not from students who skipped it. Encoding why a value is absent gave small but consistent gains over pooling all absences into one missing indicator: about 0.004 AUROC for any attempt and 0.006 for injurious attempt. The gains are modest in size. Their pattern matters more than their magnitude.

Three checks point to student-relevant information rather than survey recognition. A model using only site, year and which questions were asked barely beat chance. The gain disappeared in the national survey, where site omission does not occur. And the gain grew, rather than shrank, when whole jurisdictions were held out. This last result ran against our pre-stated hypothesis. A plausible explanation is that a single pooled indicator mixes “this survey never asks” with “this student skipped”, so its coefficient transfers poorly to a jurisdiction with a different questionnaire; separating them keeps the student-level signal from being diluted by survey design. For injurious attempt, adding site and year fixed effects removed most of the gain, so for the rarer outcome some of the advantage does track survey structure.

Multiple imputation did not help. MICE was no better than mean imputation with missing indicators (AUROC 0.841 vs 0.843 for any attempt; 0.828 vs 0.838 for injurious attempt). Imputing an item a survey never asked borrows information from other surveys and discards the fact that it was not asked. Complete-case analysis is not a realistic option in pooled YRBS data: it scored fewer than one student in eight.

At public-health screening thresholds, positive predictive values were low: at 80% sensitivity about one in four flagged students had attempted suicide, and about one in 12 to 14 had an injurious attempt. Provenance handling moved these by less than one percentage point. These models are better suited to surveillance and to studying risk structure than to

identifying individual students. Discrimination was also lower for gay or lesbian, questioning, Native Hawaiian or Pacific Islander and Black students, which deserves attention before any applied use.

The BRFSS contrast was instructive in the other direction. Most absent ACE data there came from whole states not fielding the module, and provenance handling added nothing for predicting depression. Encoding provenance helps when the absence mechanism varies within the modelling population and relates to the outcome; it is not a general fix.

Limitations. Item nonresponse, CDC edits to missing and (in national data) students who received a site questionnaire are pooled as one class, and questionnaires removed by CDC quality control are not in the release. Electronic versus paper administration can be identified only for the national survey, so mode effects on missingness were not tested at site level. Bootstrap intervals hold the fitted model fixed and do not include model-fitting variability; seed-repetition runs are planned. Bootstrap strata were collapsed to the survey level after the first run showed most design strata had one test PSU (point estimates were unaffected). Sensitivity scenarios share the primary test sets, so they are not independent replications. Four predictors matched more than one CDC variable; the choices are documented in the code release. Associations are not causal, and suicide attempts are self-reported. A branching, structure-preserving method developed with W. Merchant was not evaluated here.

Conclusion. In pooled YRBS data, telling apart “not asked” from “skipped” improves suicide-attempt prediction slightly, holds up when whole jurisdictions are held out, and is not explained by recognising surveys. Analysts pooling multi-site surveys should code the source of absence rather than treat every blank alike.

5. Usage notes

The pipeline reads the public CDC files directly (YRBS combined high-school ASCII file and SAS input program; BRFSS 2023 XPT files), so no data are redistributed. A single command runs acquisition with checksums, provenance labelling, frozen splits, model fitting and evaluation on a laptop; a Slurm chain is provided for clusters. The frozen split hashes, the variable map and the decision log are part of the release, so results can be reproduced or extended with new methods on the same test sets.

Code and data availability

YRBS combined high-school data (2023 release) and BRFSS 2023 are public-use files from CDC (YRBS data; BRFSS 2023). No restricted data were used. Analysis code, frozen split hashes, aggregate results and figures will be released on GitHub under the MIT license and archived on Zenodo after author verification.

Declarations

Funding. This work received no specific grant from any funding agency.

Ethics. This study used only de-identified, publicly available data (CDC YRBS and BRFSS public-use files) and therefore does not meet the definition of human subjects research under 45 CFR 46.102. It was not submitted for institutional review board review. No attempt was made to identify any individual.

Competing interests. None declared.

Author contributions (CRediT). Selorm Buaka: Conceptualization; Methodology; Software; Formal analysis; Writing – original draft. Sole author.

Use of AI assistance. The author conceived the study and made its design decisions

(outcomes, predictor set, provenance taxonomy, comparators and split design). An AI assistant (Claude, Anthropic) wrote the pipeline code, ran the analyses, and drafted this manuscript, including its text, figures and tables. Every number in the manuscript is generated from the released pipeline outputs.

References

1. Centers for Disease Control and Prevention. 2023 YRBS National, State, and District Combined Datasets User's Guide. Atlanta: CDC; 2024.
2. Brener ND, Mpofu JJ, Krause KH, et al. Overview and methods for the Youth Risk Behavior Surveillance System — United States, 2023. MMWR Suppl. 2024;73(4):1–12. doi:10.15585/mmwr.su7304a1
3. Sperrin M, Martin GP, Sisk R, Peek N. Missing data should be handled differently for prediction than for description or causal explanation. J Clin Epidemiol. 2020;125:183–187. doi:10.1016/j.jclinepi.2020.03.028
4. Sisk R, Sperrin M, Peek N, van Smeden M, Martin GP. Imputation and missing indicators for handling missing data in the development and deployment of clinical prediction models: a simulation study. Stat Methods Med Res. 2023. doi:10.1177/09622802231165001
5. Fletcher Mercaldo S, Blume JD. Missing data and prediction: the pattern submodel. Biostatistics. 2020;21(2):236–252. doi:10.1093/biostatistics/kxy040
6. Jung JS, et al. Prediction models for high risk of suicide in Korean adolescents using machine learning techniques. PLoS One. 2019. PMID 31170212.
7. Lim JS, et al. Prediction models for suicide attempts among adolescents using machine learning techniques. Clin Psychopharmacol Neurosci. 2022. PMID 36263637.
8. Baiden P, et al. Racial/ethnic differences in adolescents' non-response to suicide attempt: findings from the Youth Risk Behavior Survey, 2015–2019. J Psychiatr Res. 2023. PMID 37757704.
9. He Y, et al. Predicting nonresponse to sexual identity question in youth risk behavior surveillance: a machine learning analysis of complex survey data. Ann Epidemiol. 2026. PMID 41621502.
0. Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024;385:e078378. doi:10.1136/bmj-2023-078378
1. MacNell N, Feinstein L, Wilkerson J, et al. Implementing machine learning methods with complex survey data: lessons learned on the impacts of accounting sampling weights in gradient boosting. PLoS One. 2023;18(1):e0280387. doi:10.1371/journal.pone.0280387
2. Besera G, Annor FB, Swedo EA, Aslam MV, Massetti GM. Adverse childhood experiences module nonresponse: Behavioral Risk Factor Surveillance System, 2019 and 2021. Am J Prev Med. 2024;67(6):941–950.
3. Centers for Disease Control and Prevention. Behavioral Risk Factor Surveillance System 2023 Survey Data and Documentation. Atlanta: CDC; 2024.

Appendix

Table A1. Predictors and the provenance of their values, all 1,008,297 records 2015–2023 (share of records, %).

Variable	Description (CDC indicator)	Observed (%)	Not asked by survey (%)	Not on year's questionnaire (%)	Missing where asked (%)	Not applicable (%)
QN12	Carried a weapon on school property	86.7	11.6	0.0	1.7	0.0
QN14	Did not go to school because they felt unsafe	92.5	6.3	0.0	1.2	0.0
QN16	Were in a physical fight	64.0	32.5	0.0	3.5	0.0
QN19	Were ever physically forced to have sexual intercourse	70.2	27.5	0.0	2.4	0.0
QN22	Experienced physical dating violence	57.4	1.6	0.0	3.8	37.2
QN24	Were bullied on school property	87.9	10.5	0.0	1.7	0.0
QN25	Were electronically bullied	98.5	0.3	0.0	1.2	0.0
QN26	Felt sad or hopeless	98.0	0.1	0.0	1.9	0.0
QN33	Currently smoked cigarettes	94.0	2.7	0.0	3.3	0.0
QN36	Currently used electronic vapor products	90.5	2.4	0.0	7.1	0.0
QN42	Currently drank alcohol	91.1	0.7	0.0	8.2	0.0
QN48	Currently used marijuana	95.8	0.5	0.0	3.7	0.0
QN49	Ever misused prescription pain medicine	65.7	9.4	23.0	1.9	0.0
QN85	Got 8 or more hours of sleep	71.9	21.5	0.0	6.6	0.0
QN87	Described grades as mostly A's or B's	65.7	29.4	0.0	4.9	0.0

Table A2. Survey-weighted outcome prevalence by year, state and district surveys (site) and the national survey (Taylor-linearized 95% CI).

	Any past-year attempt				Injurious attempt			
Year	Site n	Site % (95% CI)	National n	National % (95% CI)	Site n	Site % (95% CI)	National n	National % (95% CI)
2015	136,020	8.8 (8.4–9.2)	12,567	8.6 (7.6–9.5)	135,432	3.2 (3.0–3.4)	12,331	2.8 (2.2–3.4)
2017	136,539	9.4 (8.9–9.9)	10,686	7.4 (6.4–8.3)	107,438	3.4 (3.1–3.6)	10,620	2.4 (2.1–2.8)
2019	143,875	9.2 (8.8–9.6)	10,520	8.9 (7.9–9.9)	113,646	3.1 (2.9–3.3)	8,749	2.5 (2.2–2.9)
2021	143,706	10.7 (10.1–11.3)	15,573	10.2 (9.4–11.0)	82,495	3.4 (3.0–3.7)	12,083	2.9 (2.5–3.4)
2023	124,548	10.2 (9.4–10.9)	19,336	9.5 (8.4–10.6)	65,696	3.2 (2.5–3.9)	14,635	2.3 (1.5–3.1)

Table A3. Pre-specified AUROC contrasts by scenario, any past-year attempt (×0.001; survey-weighted; paired 95% CI, 1,000 replicates).

Scenario	PROV_LR − MIND_LR	PATTERN_LR − MEAN_LR	PROV_HGB − HGB_NATIVE	PROV_LR − MICE_LR
Primary	+3.7 (+1.9, +5.7)	+2.7 (+1.5, +4.0)	+3.9 (+2.4, +5.3)	+5.7 (+3.0, +8.5)
Jurisdictions held out	+5.4 (+2.2, +8.6)	+4.0 (+1.2, +7.0)	+7.5 (+4.0, +11.1)	+13.5 (+6.8, +19.3)
Latest cycle held out	+4.2 (+1.7, +6.7)	+2.9 (+1.2, +4.6)	+2.8 (+0.6, +5.0)	+8.7 (+5.9, +11.5)
National survey only	−0.4 (−3.3, +1.8)	+1.7 (+0.6, +2.8)	−1.0 (−3.6, +1.0)	+5.0 (+0.6, +9.9)
Excluding 2021	+4.4 (+2.9, +6.0)	+1.9 (+0.8, +3.2)	+3.0 (+1.2, +4.7)	+6.0 (+3.3, +8.4)
Excluding 2023	+5.6 (+3.5, +7.8)	+3.1 (+1.6, +4.5)	+5.2 (+3.3, +7.4)	+7.3 (+4.3, +10.5)
2023 only	+1.6 (−1.2, +4.7)	+4.3 (−0.1, +9.6)	+3.1 (+1.0, +5.8)	+3.9 (−1.9, +9.1)
Often-omitted items dropped	+4.1 (+2.1, +6.2)	+2.2 (+1.1, +3.2)	+3.4 (+1.9, +4.8)	+5.8 (+3.1, +8.8)
95% not-asked rule	+3.7 (+1.9, +5.7)	+2.7 (+1.5, +4.0)	+3.9 (+2.4, +5.3)	+5.7 (+3.0, +8.5)
Ideation and plan added	+2.8 (+1.2, +4.7)	+2.8 (+1.2, +4.2)	+3.2 (+1.1, +5.6)	+4.1 (+1.7, +6.4)
Site and year effects added	+3.5 (+2.0, +5.2)	+1.8 (+0.5, +3.1)	+3.2 (+1.7, +4.5)	+5.3 (+3.3, +7.2)
Weighted training	+4.3 (+2.7, +6.0)	+2.4 (+0.9, +3.9)	+4.7 (+2.7, +7.0)	+6.7 (+4.0, +9.5)
Unweighted surveys included	—	—	—	—

Table A4. Pre-specified AUROC contrasts by scenario, injurious attempt (×0.001; survey-weighted; paired 95% CI, 1,000 replicates).

Scenario	PROV_LR − MIND_LR	PATTERN_LR − MEAN_LR	PROV_HGB − HGB_NATIVE	PROV_LR − MICE_LR
Primary	+5.7 (+2.3, +9.0)	+0.3 (−2.1, +2.4)	+7.6 (+3.4, +11.7)	+15.6 (+9.7, +22.3)
Jurisdictions held out	+10.4 (+3.2, +17.8)	+1.5 (−1.1, +4.1)	+17.2 (+11.3, +23.6)	+18.5 (+12.0, +25.5)
Latest cycle held out	+3.3 (−1.9, +9.5)	+1.4 (−1.6, +3.8)	+4.8 (−3.0, +13.9)	+11.0 (+6.6, +16.5)
National survey only	+2.1 (−0.1, +4.0)	+0.0 (−3.6, +4.2)	+0.2 (−4.3, +5.7)	+7.3 (+0.1, +15.2)
Excluding 2021	+6.6 (−0.7, +12.5)	−0.6 (−4.0, +2.1)	+11.9 (+7.7, +16.0)	+15.4 (+9.4, +22.0)
Excluding 2023	+5.6 (+1.3, +9.8)	+2.2 (−0.5, +4.6)	+8.0 (+2.9, +12.9)	+16.8 (+10.9, +22.7)
2023 only	−4.9 (−8.7, −1.1)	+1.4 (−8.6, +10.3)	−2.5 (−11.0, +5.6)	+6.2 (−9.0, +23.6)
Often-omitted items dropped	+6.5 (+2.7, +10.0)	+1.3 (−0.6, +3.1)	+8.7 (+5.3, +12.1)	+16.7 (+10.8, +22.8)
95% not-asked rule	+5.7 (+2.3, +9.0)	+0.3 (−2.1, +2.4)	+7.6 (+3.4, +11.7)	+15.6 (+9.7, +22.3)
Ideation and plan added	+3.7 (+0.5, +6.3)	+2.0 (+0.1, +4.1)	+5.4 (+1.2, +9.3)	+12.5 (+7.0, +18.6)
Site and year effects added	+1.8 (−2.1, +5.4)	−1.5 (−5.5, +2.9)	+2.8 (−0.5, +6.0)	+12.5 (+8.0, +17.3)
Weighted training	+5.6 (+2.3, +8.7)	−0.9 (−4.5, +2.3)	+8.1 (+3.1, +13.3)	+13.8 (+7.6, +20.1)
Unweighted surveys included	—	—	—	—

Table A5. PROV_LR discrimination by subgroup, primary test sets (survey-weighted AUROC, 95% CI). Subgroups with fewer than 30 events are not shown.

		Any past-year attempt			Injurious attempt	
Characteristic	**Level**	**Test n**	**Events**	**AUROC (95% CI)**	**Events**	**AUROC (95% CI)**
Sex	Female	69,031	7,895	0.837 (0.825–0.848)	1,693	0.849 (0.826–0.867)
	Male	66,041	4,914	0.840 (0.826–0.855)	1,272	0.829 (0.806–0.853)
Grade	9th	37,623	3,947	0.851 (0.826–0.872)	855	0.831 (0.802–0.856)
	10th	36,071	3,627	0.856 (0.837–0.871)	797	0.855 (0.833–0.875)
	11th	33,244	3,001	0.855 (0.842–0.869)	721	0.834 (0.794–0.872)
	12th	27,941	2,256	0.813 (0.789–0.832)	573	0.854 (0.826–0.883)
Race and ethnicity	American Indian or Alaska Native	2,312	373	0.814 (0.761–0.859)	101	0.872 (0.821–0.915)
	Asian	7,712	575	0.861 (0.830–0.900)	118	0.849 (0.761–0.923)
	Black or African American	15,777	1,969	0.802 (0.776–0.834)	520	0.811 (0.761–0.852)
	Hispanic or Latino	31,377	3,488	0.825 (0.806–0.841)	986	0.812 (0.777–0.842)
	Native Hawaiian or Other Pacific Islander	2,023	287	0.779 (0.703–0.851)	86	0.886 (0.821–0.930)
	White	66,544	5,043	0.862 (0.851–0.874)	914	0.860 (0.838–0.876)
	Multiple races (non-Hispanic)	7,301	930	0.862 (0.844–0.882)	199	0.870 (0.836–0.906)
Sexual identity	Heterosexual	36,553	2,574	0.851 (0.825–0.874)	336	0.841 (0.795–0.890)
	Gay or lesbian	1,928	430	0.780 (0.739–0.838)	66	0.793 (0.684–0.900)
	Bisexual	5,922	1,239	0.811 (0.777–0.848)	177	0.840 (0.812–0.873)
	Describe identity another way	2,229	485	0.838 (0.786–0.876)	80	0.855 (0.792–0.912)
	Not sure (questioning)	2,406	297	0.761 (0.696–0.823)	49	0.866 (0.807–0.904)
Survey type	District	30,155	3,048	0.829 (0.817–0.841)	883	0.816 (0.796–0.835)
	State	106,143	10,123	0.848 (0.839–0.855)	2,172	0.846 (0.828–0.864)
Survey year	2015	27,538	2,500	0.861 (0.846–0.873)	788	0.828 (0.778–0.872)
	2017	26,843	2,400	0.819 (0.789–0.845)	674	0.827 (0.796–0.853)
	2019	27,807	2,574	0.846 (0.832–0.859)	706	0.839 (0.805–0.875)
	2021	28,543	3,264	0.847 (0.831–0.862)	489	0.865 (0.829–0.901)
	2023	25,567	2,433	0.870 (0.856–0.881)	398	0.861 (0.830–0.893)

Table A6. BRFSS 2023 contrasting test bed: ever-diagnosed depressive disorder predicted from ACE items (survey-weighted, 95% CI, 1,000 replicates).

	Within-state PSU split		Leave states out	
Model	**AUROC**	**Brier**	**AUROC**	**Brier**
MEAN_LR	0.683 (0.674–0.693)	0.149 (0.146–0.152)	0.710 (0.700–0.718)	0.153 (0.150–0.156)
MIND_LR	0.683 (0.674–0.693)	0.149 (0.146–0.152)	0.709 (0.700–0.718)	0.153 (0.150–0.156)
PROV_LR	0.684 (0.675–0.693)	0.149 (0.145–0.152)	0.709 (0.700–0.718)	0.153 (0.150–0.156)
PATTERN_LR	0.685 (0.676–0.694)	0.149 (0.145–0.152)	0.709 (0.700–0.717)	0.153 (0.150–0.156)
HGB_NATIVE	0.695 (0.686–0.705)	0.147 (0.144–0.150)	0.718 (0.709–0.727)	0.151 (0.148–0.154)
PROV_HGB	0.696 (0.687–0.707)	0.147 (0.143–0.150)	0.717 (0.708–0.726)	0.151 (0.148–0.155)
DEMO_LR	0.669 (0.659–0.678)	0.151 (0.147–0.154)	0.683 (0.674–0.691)	0.158 (0.155–0.161)